Erwin T. Lau

Computational and theoretical astrophysicist: baryonic feedback in galaxy groups and clusters, and small-scale structure as a cosmological probe

Education

- Ph.D., Astronomy & Astrophysics, University of Chicago, Chicago, IL, USA, 08/2010
- M.S., Astronomy & Astrophysics, University of Chicago, Chicago, IL, USA, 03/2006
- B.S., Astronomy and General Physics, University of Michigan, Ann Arbor, MI, USA, 05/2004
- B.S.E., Engineering Physics, University of Michigan, Ann Arbor, MI, USA, 05/2004

Professional Research Experience

- Nara Women's University
  Researcher, 08/2025–present
  Sponsor: Naomi Ota
- Center for Astrophysics | Harvard & Smithsonian
  Visiting Scientist, 01/2021–07/2025
  Host: Akos Bogdán
- University of Miami, Department of Physics
  Postdoctoral Associate, 01/2018–12/2020
  Sponsor: Nico Cappelluti
- Yale University, Department of Physics, Yale Center for Astronomy & Astrophysics
  Associate Research Scientist, 09/2015–10/2017
  Postdoctoral Associate, 09/2011–08/2015
  Advisor: Daisuke Nagai
- Shanghai Astronomical Observatory, Key Laboratory for Research in Galaxies and Cosmology
  Postdoctoral Fellow, 09/2010–08/2011
  Advisor: Xiaohu Yang
- University of Chicago, Department of Astronomy & Astrophysics
  Graduate Research Assistant, 09/2005–08/2010
  Advisor: Andrey V. Kravtsov
- University of Michigan, Department of Physics
  Undergraduate Research Assistant, 09/2003–04/2004
  Advisor: Tim A. McKay

Research Interests

Computational, theoretical, and statistical modeling of galaxy clusters, groups and massive galaxies, and on their observational signatures in X-ray and microwave. Galaxy clusters and groups as cosmological probes. Physical processes in the intracluster medium. Diffuse emissions from the Intergalactic Medium and the Circumgalactic Medium. Astronomical software development.

Grants and Allocations

- *Chandra* Archival Proposal, Cycle 25, "Constraining S_8 with X-ray Angular Power Spectrum of Galaxy Clusters", USD 81k (PI)
- XSEDE Computing Research Allocation TG-AST190003, "Simulating Cosmic Weather in Galaxy Clusters", 45k node hours (PI)

Leadership

- Chair, Local Organizing Committee, "Baryons in the Universe" workshop, Nagoya, Japan, 05/2027
- Lead, simulation and theory component, XRISM Key Project proposal (led by Prof. Naomi Ota)
- Lead developer, Baryon Pasting code; Baryon Factory effort applying Baryon Pasting to multiple simulation suites (Backlight, AGORA, PINOCCHIO, Mucho-Uchuu) and to the Local Volume Complete Cluster Survey (LoVoCCS)
- Lead, CAMELS X-ray circumgalactic-medium analysis
- Sprint Coordinator, LSST Dark Energy Science Collaboration, 07/2024–10/2026

Mentoring and Teaching

Co-supervised 15 undergraduate and graduate students with Prof. Daisuke Nagai at Yale University, Prof. Nico Cappelluti at the University of Miami, and Dr. Ákos Bogdán at the Center for Astrophysics | Harvard & Smithsonian. Languages of instruction: English and Mandarin.

Graduate students: Naomi Gluck, Yale Physics (2022–present), differentiable Baryon Pasting; Han Aung, Yale Physics (2015–2017), Aung, Nagai & Lau 2021; Camille Avestruz, Yale Physics (2014–2016), Avestruz et al. 2014, 2015, 2016, now Assistant Professor, Department of Physics, University of Michigan; Kaylea Nelson, Yale Astronomy (2013–2015), Nelson et al. 2014a,b, now Director, Arts & Sciences Research Computing, Yale Center for Research Computing; Joshua Burt, Yale Physics (2014–2015); Benjamin Elder, Yale Physics (2012–2013).

Undergraduate and post-baccalaureate students: Amanda Butler Contreras, Yale (2020–2022), Butler Contreras et al. 2023; Luis Fernando Machado, Yale (2016–2017) and post-baccalaureate (2020–2021), Machado Poletti Valle et al. 2021; Mehika Patel, Yale (2022); Guanhua Chen, Miami (2019); Emil Öhman, Yale (2015–2017), ZuHone et al. 2018; Mari Kawakatsu, Yale (2016–2017); Julia Menzel, Yale (2015–2016); Christopher Cappiello, Yale (2013–2015), DeForest Pioneer Prize in Physics 2015; Maya Fishbach, Yale (2013–2015), Howard L. Schultz Prize in Physics 2015; Hendrik Kits van Heyningen, Yale (2013–2014), DeForest Pioneer Prize in Physics 2014; Dan Steinbrook, Yale (2012); Elizabeth Peng, Yale (2011–2012).

Teaching Experience

Graduate Teaching Assistant, University of Chicago, 2004–2005, 2007

Selected Invited Talks

2026/10 "Dynamics of Cosmic Structures", Zhejiang University, Hangzhou, China
2026/8 "JSPS Core-to-Core XRISM Workshop for Young Researchers", Saitama University, Saitama, Japan
2026/8 "Dynamics of Galaxy Clusters Workshop", Tokyo University of Science, Tokyo, Japan
2026/6 Nagoya University, Nagoya, Japan, Seminar
2026/1 Nara Women's University, Nara, Japan, Seminar
2025/11 Osaka University, Osaka, Japan, Seminar
2024/11 Ohio State University, Columbus, OH, CCAPP Seminar
2023/8 CMB-S4 Collaboration Meeting, Plenary Session
2022/10 University of Minnesota, Minneapolis, MN, Cosmology Seminar
2022/1 Center for Astrophysics, Cambridge, MA, High Energy Astrophysics Seminar
2021/10 Boise State University, Boise, Idaho, Computing Colloquium
2019/6 INAF-OAS, Bologna, Italy, Cluster Talk
2017/9 Yale University, CT, USA, Astronomy Colloquium
2017/2 Rochester Institute of Technology, Rochester, NY, USA, Astrophysics Colloquium
2017/1 Kavli Institute for the Physics and Mathematics of the Universe, Kashiwa, Japan, Astrophysics Seminar
2017/1 Academia Sinica Institute of Astronomy and Astrophysics, Taipei, Taiwan, Special Seminar
2016/1 Shanghai Astronomical Observatory, Shanghai, China, Special Seminar
2016/1 Kavli Institute for Astronomy and Astrophysics, Peking University, Beijing, Lunch Seminar
2015/9 Kavli Institute for the Physics and Mathematics of the Universe, Kashiwa, Japan, Astrophysics Lunch Talk
2015/3 University of Hong Kong, Hong Kong, Physics Seminar
2014/3 University of Michigan, Ann Arbor, MI, USA, Cluster Seminar
2013/10 Brown University, Providence, RI, USA, Astrophysics Seminar
2012/8 Shanghai Astronomical Observatory, Shanghai, China, Cosmology Seminar
2012/4 Yale University, New Haven, CT, USA, Cosmology Seminar
2010/3 University of Colorado, Boulder, CO, USA, Cluster Seminar

Conference and Workshop Presentations

2025/10	"XRISM International Conference 2025", Kyoto, Japan
2025/9	"CL2025: 1st East Asian Workshop on Galaxy Clusters", ASIAA, Taipei, Taiwan
2025/7	"Cosmic Ecosystems", Perimeter Institute for Theoretical Physics, Waterloo, Ontario, Canada
2025/7	"Tracing Cosmic Evolution with Galaxy Clusters V", Sexten Center for Astrophysics, Sesto, Italy
2025/6	"mm Universe 2025", University of Chicago, Chicago, Illinois, USA
2025/5	"Expanding the boundaries of dark matter halo", Shanghai Jiaotong University, Shanghai, China
2024/12	"Cosmology and Galaxy Astrophysics with Simulations & Machine Learning", Flatiron Institute, NY, USA
2024/9	"First eROSITA International Conference", MPE, Garching, Germany
2024/8	"Multiphase Madness CGM Workshop", Center for Astrophysics, Cambridge, MA
2024/7	"EAS Meeting 2024", Padova, Italy
2023/7	"CMB-S4 Summer Collaboration Meeting 2023", SLAC, Menlo Park, CA
2022/12	"CAMELS Workshop", Flatiron Institute, NY, USA
2022/6	"AGN Feeding and Feedback II", Sexten Center for Astrophysics, Sesto, Italy
2022/4	Galaxy Cluster Symposium, STSci, Baltimore, MD, USA
2019/6	XMM Workshop on Extended X-ray Sources, ESAC, Madrid, Spain
2019/1	233rd AAS meeting, Seattle, WA, USA
2018/12	"Miami 2018: Topical Physics Conference", Fort Lauderdale, FL, USA
2018/10	"Accretion Histories of AGN Workshop", Miami, FL, USA
2018/7	"Alpine Cosmology Workshop 2018", Alagna Valsesia, Italy
2018/6	"AGN Feeding and Feedback", Sexten Center for Astrophysics, Sesto, Italy
2017/8	"From Chandra to Lynx: Taking the Sharpest X-ray Vision Fainter and Farther", Cambridge, MA, USA
2016/12	"Galaxy clusters: Physics Laboratories and Cosmological Probes", Cambridge, UK
2015/6	"ICM Physics and Modelling", Max Planck Institute for Astrophysics, Garching, Germany
2015/3	"Astroparticle View of Galaxy Clusters", Hiroshima University, Hiroshima, Japan
2015/3	"SnowCluster 2015", Snowmass, UT, USA
2014/6	"Cluster Paris 2014", Paris, France
2013/3	"SnowCluster 2013", Snowmass, UT, USA
2012/11	Galaxy Cluster Workshop, Ringberg, Germany
2012/8	28th IAU General Assembly, Beijing, China
2012/6	220th AAS Meeting, Anchorage, AK, USA
2012/3	"Turbulence in Cosmic Structure Formation", ASU, Tempe, AZ, USA
2011/9	"Cosmology with X-ray and SZE Observations of Galaxy Clusters", Huntsville, AL, USA
2011/3	"Astrophysics and Cosmology with Galaxy Clusters", KITP, Santa Barbara, CA, USA
2010/7	"From Massive Galaxy Formation to Dark Energy", IPMU, Kashiwa, Chiba, Japan
2010/6	10th "Great Lakes Cosmology Workshop", Chicago, IL, USA
2010/1	215th AAS meeting, Washington, DC, USA
2007/2	"Clusters of Galaxies as Cosmological Probes", Aspen, CO, USA

Scientific Collaborations

Member, LSST Dark Energy Science Collaboration, 2017-current
External Collaborator, Euclid Collaboration, 2025-current
Provisional Member, CMB-S4 Collaboration, 2022-2025
Active Member, FornaX Collaboration
Member, OLIMPO Team
Science Working Group Member in the ICM and CGM in the Line Emission Mapper Mission
Science Working Group Member in NewAthena

Professional Activities and Academic Services

Referee for:

- Nature Communications
- The Astrophysical Journal
- The Monthly Notices of the Royal Astronomical Society
- Astronomy & Astrophysics
- Research in Astronomy & Astrophysics
- RAS Techniques and Instruments

Review Panelist, National Science Foundation - Astronomy and Astrophysics Research Grants, 2023
Peer Reviewer, *Chandra* Cycle 20 Peer Review, 2018
Organizer, Cosmology Seminar at Yale University, 2012–2014

Professional Societies

Member, American Astronomical Society
Member, International Astronomical Union

References

Prof. Naomi Ota	Nara Women's University	naomi@cc.nara-wu.ac.jp
Dr. Ákos Bogdán	Smithsonian Astrophysical Observatory	abogdan@cfa.harvard.edu
Prof. Nico Cappelluti	University of Miami	ncappelluti@miami.edu
Prof. Daisuke Nagai	Yale University	daisuke.nagai@yale.edu
Prof. Andrey V. Kravtsov	University of Chicago	kravtsov@uchicago.edu

Publications by Erwin T. Lau

Journal Publications

Summary. 62 refereed journal papers (including 1 in press), of which 14 first-author (*) and 8 student-led (†); 2 submitted papers and 1 in preparation listed separately; 3 other publications (proceedings, book chapter, white paper).
Citations: 4694; **h-index:** 35; **i10-index:** 56 (Google Scholar, October 2026).

62. * **Lau, E. T.**, Ota, N., Nagai, D., *Non-Thermal Pressure due to Gas Motions in the Intracluster Medium: Confronting XRISM/Resolve with TNG-Cluster Simulations*, 2026, PASJ, in press, arXiv:2608.04757

61. Ota, N., **Lau, E. T.**, Yamada, S., Omiya, Y., Yamaguchi, H., *Kinetic structure of the intracluster medium across nearby clusters observed with XRISM*, 2026, PASJ, 78, L39, arXiv:2607.06313

60. Saxena, H., Heinrich, A., Sayers, J., Zhuravleva, I., Bulbul, E., Sanders, J., Avestruz, C., Basu Thakur, R., Battistelli, E., Botteon, A., Cacciotti, F., Columbro, F., Coppolecchia, A., Cray, S., de Bernardis, P., De Petris, M., Lamagna, L., **Lau, E. T.**, Masi, S., Paiella, A., Piacentini, F., Rapaport, E., Rudnick, L., White, D., ZuHone, J., *Intracluster Medium Fluctuations on Scales up to 1 Mpc: A Combined eROSITA and SPT/Planck Analysis of Abell 3266*, 2026, ApJ, 1004, 15, arXiv:2602.21443

59. Cerini, G., Bellomi, E., Cappelluti, N., Khizroev, S., **Lau, E. T.**, Natarajan, P., ZuHone, J., *Revisiting galaxy cluster scaling relations through dark matter-gas coherence: scatter dependence on dynamical state*, 2025, ApJ, 991, 56, arXiv:2509.18240

58. Saxena, H., Sayers, J., Gavidia, A., Melin, J.-B., **Lau, E. T.**, Kim, J., Chappuis, L., Eckert, D., Ettori, S., Gaspari, M., Gastaldello, F., Kay, S., Lovisari, L., Oppizzi, F., Petris, M. D., Pratt, G. W., Pointecouteau, E., Rasia, E., Rossetti, M., Sereno, M., *CHEX-MATE: The Impact of Triaxiality and Orientation on Planck SZ Cluster Selection and Weak Lensing Mass Measurements*, 2025, A&A, 700, A128, arXiv:2505.23005

57. * **Lau, E. T.**, Nagai, D., Bogdán, Á., Medlock, I., Oppenheimer, B. D., Battaglia, N., Genel, S., Angles-Alcazar, D., Ni, Y., Villaescusa-Navarro, F., *X-raying CAMELS: Constraining Baryonic Feedback in the Circum-Galactic Medium with the CAMELS simulations and eRASS X-ray Observations*, 2025, ApJ, 984,190 arXiv:2412.04559

56. * **Lau, E. T.**, Bogdán, Á., Nagai, D., Cappelluti, N., Shirasaki, M., *Cosmology and Astrophysics with the Diffuse eRASS1 X-ray Angular Power Spectrum*, 2025, ApJ, 983, 8 arXiv:2410.22397

55. Hernández-Martínez, E. , Genel, S., Villaescusa-Navarro, F., Steinwandel, U. P., Lee, M. E., **Lau, E T.**, Spergel, D. N., *Cosmological and Astrophysical Parameter Inference from Stacked Galaxy Cluster Profiles Using CAMELS-zoomGZ*, 2025, ApJ, 981,170, arXiv:2410.10942

54. * **Lau, E. T.**, Nagai, D., Farahi, A., Ishiyama, T., Miyatake, H., Osato, K., Shirasaki, M., *Baryon Pasting the Uchuu Lightcone Simulation*, 2025, ApJ , 980, 122, arXiv:2411.00108

53. Shirasaki, M., Sifón, C., Miyatake, H., **Lau, E.**, Zhang, Z., Bahcall, N., Devlin, M., Dunkley, J., Farahi, A., Hilton, M., Lin, Y.-T., Nagai, D., Staggs, S. T., Sunayama, T., Spergel, D., Wollack, E. J., *Masses of Sunyaev-Zel'dovich Galaxy Clusters Detected by The Atacama Cosmology Telescope: Stacked Lensing Measurements with Subaru HSC Year 3 data*, 2024, Physical Review D, 110, 103006, arXiv:2407.08201

52. Singh, P., **Lau, E. T.** , Faerman, Y., Stern, J., Nagai, D., *Comparison of models for the warm-hot circumgalactic medium around Milky Way-like galaxies*, 2024, MNRAS, 532, 3222, arXiv:2407.06555

51. Lee, M. E., Genel, S., Wandelt, B. D., Zhang, B., Delgado, A. M., Pandey, S., **Lau, E. T.** , Carr, C., Cook, H., Nagai, D., Angles-Alcazar, D., Villaescusa-Navarro, F., Bryan, G. L. , *Zooming by in the CARPoolGP Lane: New CAMELS-TNG Simulations of Zoomed-in Massive Halos*, 2024, ApJ, 968, 11, arXiv:2403.10609

50. Zhang, Z., Farahi, A., Nagai, D., **Lau, E. T.**, Frieman, J., Ricci, M., von der Linden, A., Wu, H.-Y., and the LSST Dark Energy Science Collaboration, *Impact of Property Covariance on Cluster Weak lensing Scaling Relations*, 2024, MNRAS, 530, 3127, arXiv:2310.18266

49. Zhang, C., Zhuravleva, I., Markevitch, M., ZuHone, J., Mernier, F., Biffi, V., Bogdán, Á., Chakraborty, P., Churazov, E., Dolag, K., Ettori, S.., Forman, W. R., Jones, C., Khabibullin, I., Kilbourne, C., Kraft, R., **Lau, E. T.** , Lin, S.-C., Nagai, D., Nelson, D., Ogorzałek, A., Rasia, E., Sarkar, A., Simionescu, A., Su, Y.,Vogelsberger, M., Walker, S., *Mapping the Intracluster Medium in the Era of High-resolution X-ray Spectroscopy*, 2024, MNRAS, 530, 4234, arXiv:2310.02225

48. Bogdán, Á., Khabibullin, I., Kovács, O. E., Schellenberger, G., ZuHone, J., Burchett. J. N., Dolag. K.,, Churazov. E., Forman, W. R., Jones. C., Kilbourne, K., Kraft, R. P., **Lau, E.**, Markevitch, M., McCammon, D., Nagai, D., Nelson, D.., Ogorzałek, A., Oppenheimer, B. D., Sarkar, A., Su, Y., Truong, N., Veilleux, S., Vladutescu-Zopp. S., Zhuravleva, I., *Circumgalactic Medium on the Largest Scales: Detecting X-Ray Absorption Lines with Large-area Microcalorimeters*, 2023, ApJ, 953, 42, arXiv:2306.05449

47. † Butler Contreras, A., **Lau, E. T.**, Oppenheimer, B. D., Bogdán, A., Tillman, M., Nagai, D., Kovács, O. E., Burkhart, B., *X-ray absorption lines in the warm-hot intergalactic medium: probing Chandra observations with the CAMEL simulations*, 2023, MNRAS, 519, 2251, arXiv:2211.15675

46. * **Lau, E. T.**, Bogdán, Á., Chadayammuri, U., Nagai, D., Kraft, R., Cappelluti, N., *The X-ray Angular Power Spectrum of Extended Sources in the eROSITA Final Equatorial Depth Survey*, 2023, MNRAS, 518, 1496, arXiv:2204.13105

45. Moser, E., Battaglia, N., Nagai, D. **Lau, E.**, Machado Poletti Valle, L. F.,Villaescusa-Navarro, F., Amodeo, S., Anglés-Alcázar, D., Bryan, G. L., Davé, R., Hernquist, L., Vogelsberger, M., *The Circumgalactic Medium from the CAMELS Simulations: Forecasting Constraints on Feedback Processes from Future Sunyaev-Zeldovich Observations*, 2022, ApJ, 933, 133, arXiv:2201.02708

44. Villaescusa-Navarro, F., Genel, S., Angles-Alcazar, D., Thiele, L., Davé, R., Narayanan, D., Nicola, A., Li, Y., Villanueva-Domingo, P., Wandelt, B., Spergel, D. N., Somerville, R. S., Zorrilla M., José, M., Mohammad, F. G., Hassan, S., Shao, H., Wadekar, D., Eickenberg, M., Wong, K. W.-K., Contardo, G., Jo, Y., Moser, E., **Lau, E. T.**, Machado Poletti Valle, L. F., Perez, L. A., Nagai, D., Battaglia, N., Vogelsberger, M., *The CAMELS Multifield Dataset: Learning the Universe's Fundamental Parameters with Artificial Intelligence*, 2022, ApJS, 259, 61, arXiv:2109.10915

43. Stapelberg, S., Tchernin, C., Hug, D., **Lau, E. T.**, Bartelmann, M., *Triaxiality in galaxy clusters: Mass versus Potential reconstructions*, 2021 A&A, 663, A17 arXiv:2012.13413

42. † Aung, H., Nagai, D., **Lau, E. T.**, *Shock and Splash: Gas and Dark Matter Halo Boundaries around ΛCDM Galaxy Clusters*, 2021, MNRAS, 508, 2071, arXiv:2012.00977

41. † Machado Poletti Valle, L. F., Avestruz, C., Barnes, D. J., Farahi, A., **Lau, E. T.**, Nagai, D., *SHAPing the Gas: Understanding Gas Shapes in Dark Matter Haloes with Interpretable Machine Learning*, 2021, MNRAS, 507, 1468 arXiv:2011.12987

40. * **Lau, E. T.**, Hearin, A. P., Nagai, D., Cappelluti, N., *Correlations between Triaxial Shapes and Formation History of Dark Matter Haloes*, 2021, MNRAS, 500, 1029, arXiv:2006.09420

39. Simionescu, A., Ettori, S., Werner, N., Nagai, D., Vazza, F., Akamatsu, H., Pinto, C., de Plaa, J., Wijers, N., Nelson, D., Pointecouteau, E., Pratt, G. W., Spiga, D., **Lau, E.**, Rossetti, M., Gastaldello, F., Biffi, V., Bulbul, E., den Herder, J. W., Eckert, D. Fraternali, F., Mingo, B., Pareschi, G., Pezzulli, G., Reiprich, T. H., Schaye, J., Walker, S. A., Werk, J., *Voyage through the Hidden Physics of the Cosmic Web*, 2021, Experimental Astronomy, 51, 1043, arXiv:1908.01778

38. Comparat, J., Eckert, D., Finoguenov, A., Schmidt, R., Sanders, J., Nagai, D., **Lau, E. T.**, Kaefer, F., Pacaud, F., Clerc, N., Reiprich, T. H., Bulbul, E., Ider Chitham, J., Chuang, C.-H., Ghirardini, V., Gonzalez-Perez, V., Gozaliazl, G.,

Kirkpatrick, C. C., Klypin, A., Merloni, A., Nandra, K., Liu, T., Prada, F., Ramos-Ceja, M. E., Salvato, M., Seppi, R., Tempel, E., Yepes, G., *Full-sky photon simulation of clusters and active galactic nuclei in the soft X-rays for eROSITA*, 2020, The Open Journal of Astrophysics, 3, 13, arXiv:2008.08404

37. Tchernin, C., **Lau, E. T.**, Stapelberg, S., Hug, D., Bartelmann, M., *Characterizing galaxy clusters by their gravitational potential: systematics of cluster potential reconstruction*, 2020, A&A, 644, A126, arXiv:2008.01107

36. Shirasaki, M., **Lau, E. T.**, Nagai, D., *Probing Cosmology and Cluster Astrophysics with Multi-Wavelength Surveys I. Correlation Statistics*, 2020, MNRAS, 491, 235, arXiv:1909.02179

35. Chen, H., Avestruz, C., Kravtsov, A V., **Lau, E. T.**, Nagai, D., *Imprints of mass accretion history on the shape of the intracluster medium and the $T_X - M$ relation*, 2019, MNRAS, 490, 2380 arXiv:1903.08662

34. Shi, X., Nagai, D., **Lau, E. T.**, *Multiscale analysis of turbulence evolution in the density-stratified intracluster medium*, 2018, MNRAS, 481, 1075 arXiv:1806.05056

33. Shirasaki, M., **Lau, E. T.**, Nagai, D., *Modelling Baryonic Effects on Galaxy Cluster Mass Profiles*, 2018, MNRAS, 477, 2804, arXiv:1711.06366

32. ZuHone, J. A., Kowalik, K., Öhman, E., **Lau, E.**, Nagai, D., *The Galaxy Cluster Merger Catalog: An Online Repository of Mock Observations from Simulated Galaxy Cluster Mergers*, 2018, ApJS, 234, 4, arXiv:1609.04121

31. Zinger, E., Dekel, A., Birnboim, Y., Nagai, D., **Lau, E.**, Kravtsov, A. V., *Cold Fronts and Shocks Formed by Gas Streams in Galaxy Clusters*, 2018, MNRAS, 476, 56, arXiv:1609.05308

30. Ota, N., Nagai, D., **Lau, E. T.**, *Constraining hydrostatic mass bias of galaxy clusters with high-resolution X-ray spectroscopy*, 2018, PASJ, 70, 51, arXiv:1507.02730

29. * **Lau, E. T.**, Gaspari, M., Nagai, D., Coppi, P., *Physical Origins of Gas Motions in Galaxy Cluster Cores: Interpreting Hitomi Observations of the Perseus Cluster*, 2017, ApJ, 849, 54, arXiv:1705.06280

28. Tchernin, C., Eckert, D., Ettori, S., Pointecouteau, E.. Paltani, S., Molendi, S., Hurier, G., Gastaldello, F., **Lau, E. T.**, Nagai, D., Roncarelli, M.. Rossetti, M., *The XMM Cluster Outskirts Project (X-COP): Physical conditions to the virial radius of Abell 2142*, 2016, A&A, 595, A42, arXiv:1606.05657

27. † Avestruz, C., Nagai, D., **Lau, E. T.**, *Stirred, not Clumped: Evolution of Temperature Profiles in the Outskirts of Galaxy Clusters*, 2016, ApJ, 833, 227, arXiv:1605.01723

26. Shirasaki, M., Nagai, D., **Lau, E. T.**, *Covariance in the Thermal SZ-Weak Lensing Mass Scaling Relation of Galaxy Clusters*, 2016, MNRAS, 460, 3913, arXiv:1603.08609

25. Shi, X., Komatsu, E., Nagai, D., **Lau, E. T.**, *Analytical model for non-thermal pressure in galaxy clusters - III. Removing the hydrostatic mass bias*, 2016, MNRAS, 455, 2936, arXiv:1507.04338

24. Sembolini, F. , Yepes, G., Pearce, F. R., Knebe, A., Kay, S. T., Power, C., Cui, W., Beck, A. M., Borgani, S., Dalla Vecchia, C., Davé, R., Elahi, P. J., February, S., Huang, S., Hobbs, A., Katz, N., **Lau, E.**, McCarthy, I. G., Murante, G., Nagai, D., Nelson, K., Newton, R. D. A., Perret, V., Puchwein, E., Read, J. I., Saro, A., Schaye, J., Teyssier, R.,Thacker, R. J., *nIFTy galaxy cluster simulations - I. Dark matter and non-radiative models*, 2016, MNRAS, 457, 2063, arXiv:1503.06065

23. * **Lau, E. T.**, Nagai, D., Avestruz, C., Nelson, K., Vikhlinin, A., *Mass Accretion and its Effects on the Self-similarity of Gas Profiles in the Outskirts of Galaxy Clusters*, 2015, ApJ, 806, 86, arXiv:1411.5361

22. † Avestruz, C., Nagai, D., **Lau, E. T.**, Nelson, K., *Non-Equilibrium Electrons in the Outskirts of Galaxy Clusters*, 2015, ApJ, 808, 176, arXiv:1410.8142

21. Rasia, E., **Lau, E. T.**, Borgani, S., Nagai, D., Dolag K., Avestruz, C., Granato, G. L., Mazzotta, P., Murante, G., Nelson, K., Ragone-Figueroa, C., *Temperature Structure of the Intra-Cluster Medium from SPH and AMR simulation*, 2014, ApJ, 791, 96, arXiv:1406.4410

20. Zhuravleva, I., Churazov, E., Schekochihin, A. A., **Lau, E. T.**, Nagai, D., Gaspari, M., Allen, S. W., Nelson, K., Parrish, I. J. *The relation between gas density and velocity power spectra in galaxy clusters: qualitative treatment and cosmological simulations* 2014, ApJL, 788, L13, arXiv:1404.5306

19. Gaspari, M., Churazov, E., Nagai, D., **Lau, E. T.**, Zhuravleva, I., *The relation between gas density and velocity power spectra in galaxy clusters: high-resolution hydrodynamic simulations and the role of conduction*, 2014, A&A, 569, A67, arXiv:1404.5302

18. † Nelson, K., **Lau, E. T.**, Nagai, D., *Hydrodynamic Simulation of Non-thermal Pressure Profiles of Galaxy Clusters*, 2014, ApJ, 792, 25, arXiv:1404.4636

17. † Avestruz, C., **Lau, E. T.**, Nagai, D., Vikhlinin, A., *Testing X-ray Measurements of Galaxy Cluster Outskirts with Cosmological Simulations*, 2014, ApJ, 797, 117, arXiv:1404.4634

16. Wang, L., Yang, X., Shen, S., Mo, H. J., van den Bosch., F. C., Luo, W., Wang, Y., **Lau, E. T.**, Wang, Q. D., Kang, X., Li, R., *Measuring the X-ray luminosities of SDSS DR7 clusters from RASS*, 2014, MNRAS, 439, 611, arXiv:1312.7417

15. Harvey, D., Tittley, E., Massey, R., Kitching, T. D., Taylor, A., Pike, S. R., Kay, S. .T., **Lau, E. T.**, & Nagai, D., *On the cross-section of Dark Matter using substructure infall into galaxy clusters*, 2014, MNRAS, 441, 404, arXiv:1310.1731

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